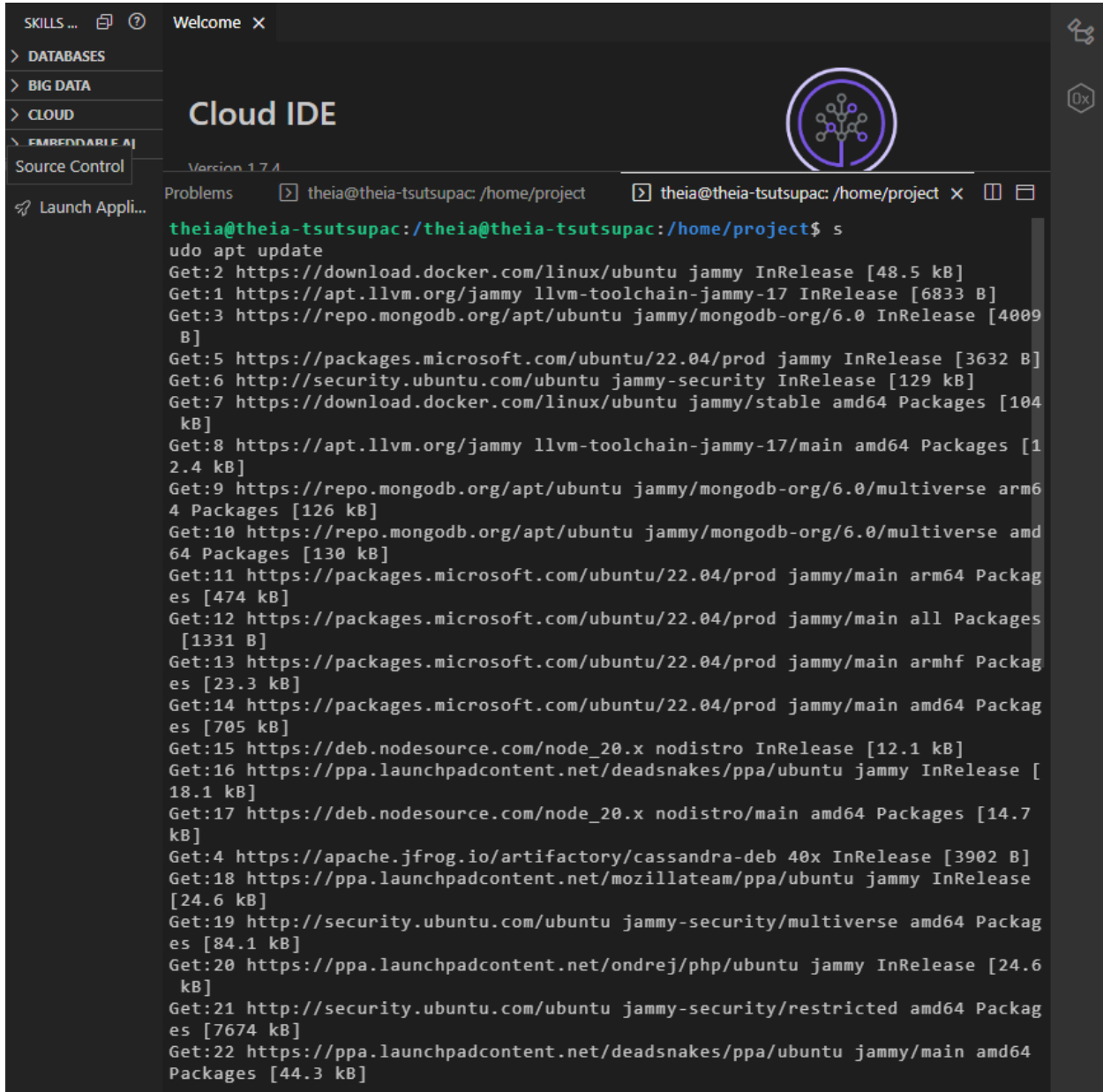


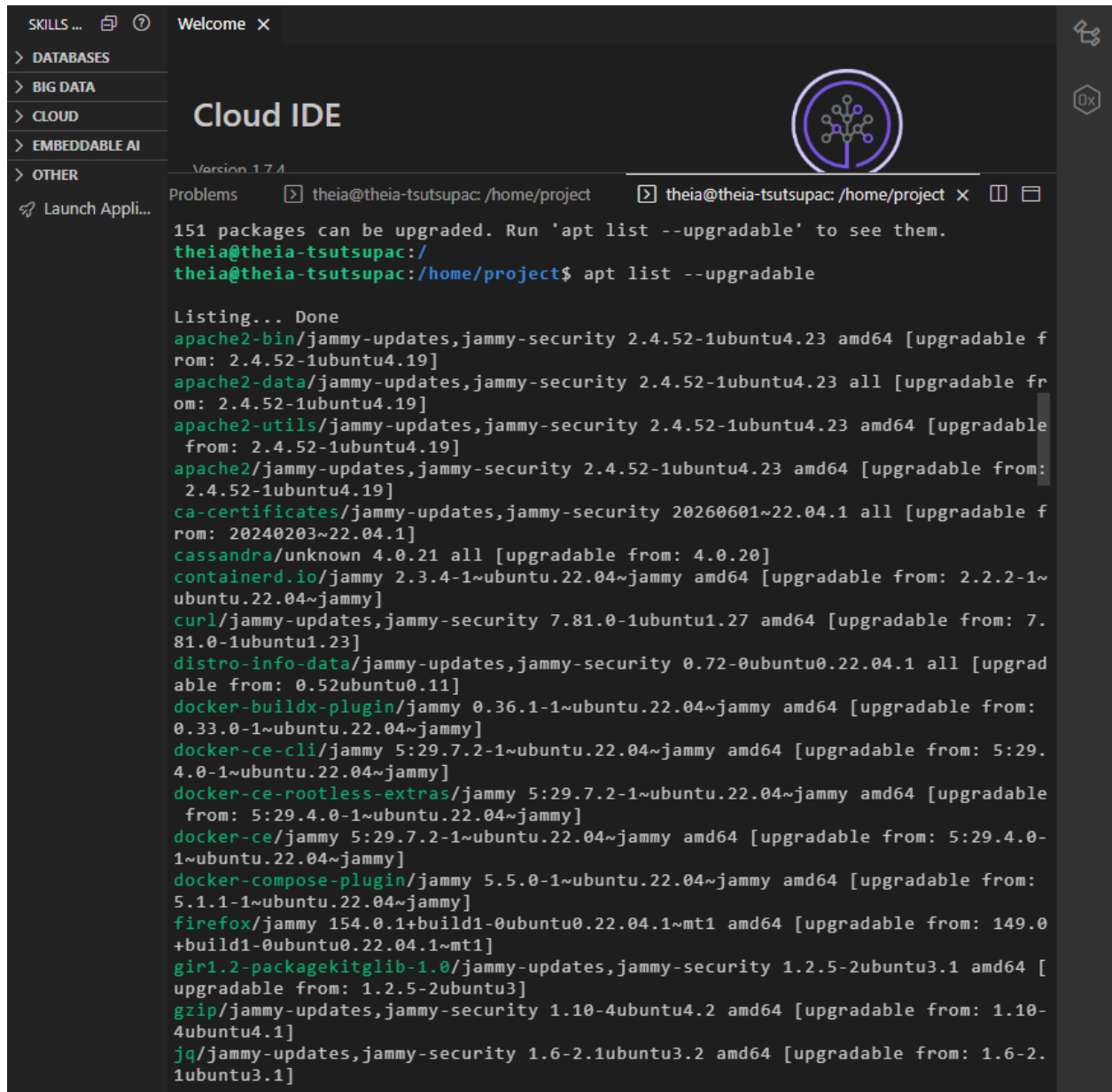
## Exercise 1: Update Package Lists

- 1.) “Apt update” is used to update the local package index with the latest changes made in the repositories. This command retrieves information about available package versions from the system’s configured repositories. If there is a new version, the system marks it for download and installation. The command does not upgrade or install any packages.

The screenshot shows the Cloud IDE interface with a dark theme. On the left, there is a sidebar with navigation options: SKILLS, DATABASES, BIG DATA, CLOUD, EMBEDDABLE AI, and Source Control. The main area is titled 'Cloud IDE' and 'Version 1.7.4'. Below this, there are tabs for 'Problems' and two terminal windows. The active terminal window shows the command 'sudo apt update' being executed. The output lists various repositories and the packages they provide, including Docker, LLVM, MongoDB, Microsoft, Ubuntu Security, and NodeSource. The terminal output is as follows:

```
theia@theia-tsutsupac:/theia@theia-tsutsupac:/home/project$ sudo apt update
Get:2 https://download.docker.com/linux/ubuntu jammy InRelease [48.5 kB]
Get:1 https://apt.llvm.org/jammy llvm-toolchain-jammy-17 InRelease [6833 B]
Get:3 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 InRelease [4009 B]
Get:5 https://packages.microsoft.com/ubuntu/22.04/prod jammy InRelease [3632 B]
Get:6 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:7 https://download.docker.com/linux/ubuntu jammy/stable amd64 Packages [104 kB]
Get:8 https://apt.llvm.org/jammy llvm-toolchain-jammy-17/main amd64 Packages [12.4 kB]
Get:9 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0/multiverse arm64 Packages [126 kB]
Get:10 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0/multiverse amd64 Packages [130 kB]
Get:11 https://packages.microsoft.com/ubuntu/22.04/prod jammy/main arm64 Packages [474 kB]
Get:12 https://packages.microsoft.com/ubuntu/22.04/prod jammy/main all Packages [1331 B]
Get:13 https://packages.microsoft.com/ubuntu/22.04/prod jammy/main armhf Packages [23.3 kB]
Get:14 https://packages.microsoft.com/ubuntu/22.04/prod jammy/main amd64 Packages [705 kB]
Get:15 https://deb.nodesource.com/node_20.x nodistro InRelease [12.1 kB]
Get:16 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease [18.1 kB]
Get:17 https://deb.nodesource.com/node_20.x nodistro/main amd64 Packages [14.7 kB]
Get:4 https://apache.jfrog.io/artifactory/cassandra-deb 40x InRelease [3902 B]
Get:18 https://ppa.launchpadcontent.net/mozillateam/ppa/ubuntu jammy InRelease [24.6 kB]
Get:19 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [84.1 kB]
Get:20 https://ppa.launchpadcontent.net/ondrej/php/ubuntu jammy InRelease [24.6 kB]
Get:21 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [7674 kB]
Get:22 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64 Packages [44.3 kB]
```

- 2.) “Apt list –upgradable” tells you which packages are available for upgrade



The screenshot shows the Cloud IDE interface. On the left is a sidebar with a menu: SKILLS ... (with a folder icon), Welcome X, > DATABASES, > BIG DATA, > CLOUD, > EMBEDDABLE AI, > OTHER, and a Launch Appli... button. The main area has a header with 'Cloud IDE' and 'Version 1.7.4'. Below the header is a terminal window with the title 'theia@theia-tsutsupac: /home/project'. The terminal output shows a message about 151 upgradable packages, followed by the command 'apt list --upgradable' and its output, which lists various packages and their upgrade paths. The packages listed include apache2-bin, apache2-data, apache2-utils, apache2, ca-certificates, cassandra, containerd.io, curl, distro-info-data, docker-buildx-plugin, docker-ce-cli, docker-ce-rootless-extras, docker-ce, docker-compose-plugin, firefox, gir1.2-packagekitglib-1.0, gzip, and jq.

```
151 packages can be upgraded. Run 'apt list --upgradable' to see them.
theia@theia-tsutsupac:/
theia@theia-tsutsupac:/home/project$ apt list --upgradable

Listing... Done
apache2-bin/jammy-updates,jammy-security 2.4.52-1ubuntu4.23 amd64 [upgradable from: 2.4.52-1ubuntu4.19]
apache2-data/jammy-updates,jammy-security 2.4.52-1ubuntu4.23 all [upgradable from: 2.4.52-1ubuntu4.19]
apache2-utils/jammy-updates,jammy-security 2.4.52-1ubuntu4.23 amd64 [upgradable from: 2.4.52-1ubuntu4.19]
apache2/jammy-updates,jammy-security 2.4.52-1ubuntu4.23 amd64 [upgradable from: 2.4.52-1ubuntu4.19]
ca-certificates/jammy-updates,jammy-security 20240203~22.04.1 all [upgradable from: 20240203~22.04.1]
cassandra/unknown 4.0.21 all [upgradable from: 4.0.20]
containerd.io/jammy 2.3.4-1ubuntu.22.04~jammy amd64 [upgradable from: 2.2.2-1ubuntu.22.04~jammy]
curl/jammy-updates,jammy-security 7.81.0-1ubuntu1.27 amd64 [upgradable from: 7.81.0-1ubuntu1.23]
distro-info-data/jammy-updates,jammy-security 0.72-0ubuntu0.22.04.1 all [upgradable from: 0.52ubuntu0.11]
docker-buildx-plugin/jammy 0.36.1-1ubuntu.22.04~jammy amd64 [upgradable from: 0.33.0-1ubuntu.22.04~jammy]
docker-ce-cli/jammy 5:29.7.2-1ubuntu.22.04~jammy amd64 [upgradable from: 5:29.4.0-1ubuntu.22.04~jammy]
docker-ce-rootless-extras/jammy 5:29.7.2-1ubuntu.22.04~jammy amd64 [upgradable from: 5:29.4.0-1ubuntu.22.04~jammy]
docker-ce/jammy 5:29.7.2-1ubuntu.22.04~jammy amd64 [upgradable from: 5:29.4.0-1ubuntu.22.04~jammy]
docker-compose-plugin/jammy 5.5.0-1ubuntu.22.04~jammy amd64 [upgradable from: 5.1.1-1ubuntu.22.04~jammy]
firefox/jammy 154.0.1+build1-0ubuntu0.22.04.1~mt1 amd64 [upgradable from: 149.0+build1-0ubuntu0.22.04.1~mt1]
gir1.2-packagekitglib-1.0/jammy-updates,jammy-security 1.2.5-2ubuntu3.1 amd64 [upgradable from: 1.2.5-2ubuntu3]
gzip/jammy-updates,jammy-security 1.10-4ubuntu4.2 amd64 [upgradable from: 1.10-4ubuntu4.1]
jq/jammy-updates,jammy-security 1.6-2.1ubuntu3.2 amd64 [upgradable from: 1.6-2.1ubuntu3.1]
```

## Exercise 2: Perform System Upgrade

- 1.) “Sudo apt upgrade” used to upgrade the installed packages and the Linux kernel to the latest versions.

SKILLS ...

Welcome X

> DATABASES

> BIG DATA

> CLOUD

> EMBEDDABLE AI

> OTHER

Launch Appli...

Cloud IDE

Version 1.7.4



Problems

theia@theia-tsutsupac: /home/project

theia@theia-tsutsupac: /home/project X

```
invoke-rc.d: policy-rc.d denied execution of restart.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of restart.
Setting up openjdk-11-jre-headless:amd64 (11.0.32+9-1ubuntu1~22.04) ...
Installing new version of config file /etc/java-11-openjdk/jfr/default.jfc ...
Installing new version of config file /etc/java-11-openjdk/jfr/profile.jfc ...
Installing new version of config file /etc/java-11-openjdk/security/java.securi
ty ...
Setting up pkexec (0.105-33ubuntu0.1) ...
Setting up cassandra (4.0.21) ...
sysctl: setting key "vm.max_map_count", ignoring: Read-only file system
sysctl: setting key "net.ipv4.tcp_keepalive_time", ignoring: Read-only file sys
tem
Setting up php8.1 (8.1.34-8+ubuntu22.04.1+deb.sury.org+1) ...
Setting up policykit-1 (0.105-33ubuntu0.1) ...
Setting up packagekit (1.2.5-2ubuntu3.1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of force-reload.
Failed to open connection to "system" message bus: Failed to connect to socket
/run/dbus/system_bus_socket: No such file or directory
Setting up packagekit-tools (1.2.5-2ubuntu3.1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.14) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for dbus (1.12.20-2ubuntu4.1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for ca-certificates (20260601~22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...

done.
done.
Processing triggers for php8.1-cli (8.1.34-8+ubuntu22.04.1+deb.sury.org+1) ...
Processing triggers for libapache2-mod-php8.1 (8.1.34-8+ubuntu22.04.1+deb.sury.
org+1) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of restart.
theia@theia-tsutsupac:/home/project$
```